

SI Session 3

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February 2026

1 Mixture Rate Problem

2 Problem 1

A 600-liter tank initially contains 79 liters of a fluid in which 223 grams of a certain chemical have been dissolved. Starting at $t = 0$, fluid containing 30 grams per liter of the same dissolved chemical flows into the tank at 6 liters/min. If the mixture is kept uniform by constant stirring and if the well-stirred mixture flows out of the tank at 3 liters/min, (a) how much of the chemical is in the tank at the end of 10 minutes, and (b) how much of the chemical is in the tank when the tank overflows?

2.1 Solution

$$\frac{dx}{dt} = 180 - \frac{3x}{79 + 3t}$$

$$\frac{dx}{dt} + \frac{3x}{79 + 3t} = 180$$

$$IF = e^{\int \frac{3}{79+3t} dt} = 79 + 3t$$

$$x = 30(79 + 3t) + \frac{K_1}{79 + 3t}$$

$$K_1 = (223 - 30(79))(79)$$

Part (a)

$$x = 30(79 + 30) - \frac{K_1}{79 + 30}$$

Part (b)

$$x = 30(79 + 3(\frac{521}{3})) - \frac{K_1}{79 + 3(\frac{521}{3})}$$

3 Problem 2

At $t = 0$, a 16 pound object is pushed beneath the surface of a calm lake, and then released from rest. Since the object has a buoyancy force of 32 pounds, the object begins to rise after it is released. The resistance (in pounds) of the water is $v/4$, where v is the velocity (in feet/sec) of the rising object. If the object reaches the surface in 8 seconds, what is the velocity of the object when it reaches the surface of the lake?

3.1 Solution

$$\frac{1}{2} \frac{dv}{dt} = 32 - 16 - \frac{v}{4}$$

$$\frac{dv}{dt} + \frac{v}{2} = 32$$

$$IF = e^{\int \frac{1}{2} dt} = e^{\frac{t}{2}}$$

$$v = 64 + K_1 e^{-\frac{t}{2}}$$

$$K_1 = 0 - 64$$

$$v = 64 + K_1 e^{-\frac{t}{2}}$$